

Aditya Sahni

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EDUCATION

McMaster University

Bachelor of Applied Science in Honors Computer Science with a Minor in Statistics

Hamilton, Ontario

Aug. 2022 – July 2027

EXPERIENCE

TD Bank

Software Engineer Co-op

Toronto, Ontario

Sept 2025 – Dec 2025

- Engineered an automated ETL pipeline using **Power Automate** to extract unstructured data from Confluence and transform it into SQL-ready CSV formats, reducing manual data entry latency by 90%.
- Developed complex **SQL stored procedures** to process TMX user requests, automatically categorizing users into Blocklist, Watchlist, and Safelist based on real-time security logic.
- Eliminated **5+ hours of weekly** manual processing by implementing validation scripts that ensure data integrity prior to ingestion.
- Designed a high-availability **Power BI dashboard** to monitor aggregator login trends and SLA metrics; implemented DAX optimization to reduce report refresh time and enhance query performance.

DocOnline Health India Pvt. Ltd.

Assistant Software Developer

Mumbai, India

May 2024 – Sep 2024

- Collaborated in an Agile team to build a web platform for medical data aggregation and engineered **REST APIs** for real-time validation.
- Architected a cost-efficient serverless solution using **AWS EC2, S3, and Lambda**, reducing operational infrastructure overhead.

Axesouár — Jewellery (Freelance)

Full-Stack Web Developer

Hamilton, Ontario

Jan 2025 – March 2025

- Built a responsive e-commerce platform using **React.js** and a **Node.js/MongoDB** backend to manage 120+ unique SKUs.
- Optimized queries and assets, resulting in a **60% improvement in page load speeds** and increased user session retention.

PROJECTS

Flux — Institutional-Grade Trading Platform | *Go, PostgreSQL, Redis, Next.js, Docker*

- Architected a distributed microservices trading engine in **Go (Golang)**, utilizing **goroutine-based concurrency** and **mutex locking** to aggregate disparate market streams into a unified **WebSocket** feed, achieving sub-second data synchronization.
- Implemented a financial-grade ledger system using **PostgreSQL** transactions with **row-level locking** (SELECT FOR UPDATE), ensuring strict **ACID compliance** to prevent race conditions and guarantee portfolio accuracy.
- Designed a high-performance **Next.js 14** dashboard featuring **GPU-accelerated** candlestick charts (via HTML5 Canvas) and optimistic UI updates, delivering a fluid 60fps experience for real-time market visualization.
- Deployed a **hybrid data strategy** utilizing **Redis** for hot-data caching and **Docker** for containerized consistency, reducing infrastructure overhead while ensuring high availability across development and production environments.

Emotion Detector Using Deep Learning | *TensorFlow, Keras, OpenCV, Flask*

- Developed a Convolutional Neural Network (CNN) to detect facial emotions, achieving high accuracy on the **FER-2013 dataset**.
- Implemented real-time computer vision capabilities using **OpenCV** and exposed the model via a **Flask API** for web integration.

Personal Portfolio Website | *React.js, Next.js, TailwindCSS, Vercel*

- Developed a performant, SEO-optimized portfolio using **Next.js**, **Tailwind CSS**, and **Framer Motion** for a responsive UI.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML/CSS, Go (Golang)

Frameworks: Node.js, React.js, Next.js, Flask, TensorFlow, Keras, Chart.js, Pandas, Gin

Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, Database Systems, Computer Networks, Web Dev

Tools: AWS (EC2, S3, Lambda), Docker, Git, Linux, MongoDB, PostgreSQL, VS Code, Redis, WebSockets